MILITARY SPECIFICATION

GRENADE, HE, G.P., M46
LOADING AND ASSEMBLING

This Amendment forms a part of Military Specification MIL-G-48062C (AR), dated 4 January 1982, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1: Under US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER, add the following:

"PRODUCT AND PACKAGING DRAWINGS

9349337 - Packing and Marking for Box Outer for Grenade: M42, M46 and XM77."

PAGE 3

* 3.2: Add "(see 6.12)" at end of paragraph.

3.2.1: Add the following new paragraph:

"3.2.1 Lead cup assembly. The lead cup assembly shall be free of all splits or cracks."

* Delete paragraph 3.3.1 and substitute:

"3.3.1 Normal standoff.

3.3.1.1 Cone configuration (Dwg. 9215347). When tested at normal standoff, the body loading assembly, (Dwg. 9215372) shall penetrate a minimum of 3.00 inches of mild steel or a minimum of 2.50 inches of ammunition testing plate.

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3.3.2.1 Liner configuration (Dwg. 9349371). When tested at normal standoff, the body loading assembly, (Dwg. 9215371) shall penetrate a minimum of 4.00 inches of mild steel or a minimum of 3.25 inches of ammunition testing plate."

* Delete paragraph 3.3.2 and substitute:

"3.3.2 Modified standoff.

3.3.2.1 Cone configuration. When tested at a standoff distance, which is 4 1/4 + 1/16 inches greater than the normal standoff distance, the body loading assembly shall penetrate a minimum of 5.50 inches of mild steel or a minimum of 4.50 inches of ammunition testing plate.

3.3.2.2 Liner configuration. When tested at a standoff distance, which is 3 1/2 + 1/16 inches greater than the normal standoff distance, the body loading assembly shall penetrate a minimum of 7.00 inches of mild steel (2 blocks - 5.50 and 1.50 inches) or a minimum of 5.5 inches of ammunition testing plate (2 blocks - 4.50 and 1.00 inches)."

PAGE 4

4.3.1: Add the following first article sample items:

"Packable slider lock 9329680 30"

PAGE 5

4.3.2, Table I: Under the heading of Body Loading Assy, delete "3.2.1" and substitute "3.2" and delete "4.5.5.1" and substitute "4.5.5" for unseating force of cone swage. Delete "4.5.5.2 or 4.5.5.3" and substitute "4.5.5" for test for presence of caked explosive entrapped in groove.

Add the following inspection:

"Packable slider lock (Dwg. 9329680), Examination for defects per 4.4.2.9 30 3.2".

PAGE 6

* Delete paragraph 4.3.3 and substitute:

"4.3.3 Rejection. See MIL-A-48078(AR).

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4.3.3.1 Cone configuration. In addition, if any body fuze assembly fails the normal standoff static penetration test, or if two (2) or more body fuze assemblies fail the modified standoff static penetration test, or if two (2) or more samples fail the functioning test, the first article sample shall be rejected.

4.3.3.2 Liner configuration. In addition, if any body fuze assembly fails the normal standoff static penetration test, or if the sample (30) tested at a modified standoff fails to meet the accept/reject criteria as specified in paragraph 4.4.3.5, or if two (2) samples fail the functioning test, the first article sample shall be rejected."

PAGE 10

4.4.2.4, Under Critical category: Add the following:

"2. Lead cup assembly split or cracked 100% 3.2.1 Visual".

Under Majors 101 and 102: Delete "4.5.5.1" and "4.5.5.2/4.5.5.3" and substitute "4.5.5".

* Under Major 110 insert "(Cone), 4.4.3.5 (liner)" after note b.

* Note a: Delete in its entirety and substitute:

"For each test randomly select 50 samples per lot or select 2 samples per machine at the start of production each day and 2 samples per machine after every 4 hours of production or portion thereof. Accept no failures and reject on 1 failure."

PAGE 13

4.4.2.6: Add Critical Defects 3 and 4 as follows:

"3. Slider lock cracked, incomplete, missing, or improperly positioned 100% 3.2 Visual

4. Safety clip of fuze missing, damaged, or double end not protruding beyond end of fuze tunnel. Note (c) 100% 3.2 Visual".

Delete Major Defect 103.

* Major Defects 104 and 107: Add "Note e":

"Note e: Requires slider lock removal if slider lock is automatically assembled during the fuze assembly operation."

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4.4.2.6: Add Major Defects 114, 115, 116 and 117 as follows:

"114. Height from the shoulder of grenade body to bottom of wound ribbon (two places) .40% 3.2 Gage

* 115. Deleted
* 116. Deleted
* 117. Deleted

Add Note c:

"Note c: Applies only if slide lock is not assembled immediately after tape winding operation.

* Note d: Deleted

Add the following new tables: 4.4.2.7; 4.4.2.8; 4.4.2.9

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Packing and Marking for Box, Fiberboard for Grenade, GP: M42, M46 and XM77	1 OF 1	9349336
			NEXT HIGHER ASSEMBLY 9349337

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Partitions missing		0.40%	3.2	Visual
102	Tubes missing		0.40%	3.2	Visual
103	Barrier bag torn, punctured or heat seal incomplete		0.40%	3.2	Visual
104	Box damaged, exposing contents		0.40%	3.2	Visual
105	Corners not blunted		0.40%	3.2	Visual
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Box marking missing, incomplete or illegible		0.65%	3.2	Visual
203	Barrier bag marking missing, incomplete or illegible		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9349337
4.4.2.8	Packing and Marking for Box, Outer for Grenade: M42, M46 and XM77				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Contents loose		0.40%	3.2	Manual
102	Box damaged, exposing contents		0.40%	3.2	Visual
103	Strapping missing (wood box only)		0.40%	3.2	Visual
104	Seal missing (wire bound box only)		0.40%	3.2	Visual
Minor					
201	Marking missing, incomplete or illegible		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Packable Slider Lock		9329680
			NEXT HIGHER ASSEMBLY 9215370

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Locking tab cracked, incomplete or missing		100%	3.2	Visual
Major					
101	Profile of minimum interior dimensions undersize		0.40%	3.2	Gage
102	Location of locking tab		0.40%	3.2	Gage
103	Length of locking tab		0.40%	3.2	Gage
104	Location of safety pin hole		0.40%	3.2	Gage
105	Width of safety pin hole		0.40%	3.2	Gage
106	Length of safety pin hole		0.40%	3.2	Gage
107	Pull test, max.		0.40%	3.2	4.5.9
108	Pull test, min.	(a)		3.2	4.5.9
109	Hardness test	(b or c)		3.2	4.5.10
Minor					
201	Width		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Protective coating incomplete or missing		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
(a) The same sample from Major 107, accept on 0 - reject on 1.
(b) 25 samples per 4 hours production, accept on 0 - reject on 1, by the producer.
(c) When alternate material (Martensite steel) is used, apply special inspection level S-4, AQL 0.15%. Manufacturer's certification of hardness for preheat treated material is required.

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PAGE 16

Add the following new paragraph 4.4.3.5:

"4.4.3.5 Static penetration test of body loading assembly (modified standoff) major defect. After completion of firings for each block assembly, an observation shall be made first of the total number of penetrations of the 7.00 inch block assembly and then the total number of penetrations of the 5.50 inch block of mild steel. Similarly, if the ammunition test plate is used, the number of penetration of the 5.50 inch block assembly shall be noted first and then the total number of penetrations of the 4.50 inch block. The test sample fired (30) shall meet the following accept/reject criteria:

a. Mild steel. Accept on 2 failures, reject on 3 failures for the 5.50 inch block and accept on 19 failures, reject on 20 failures for the 7.00 inch block assembly.

b. Ammunition test plate. Accept on 2 failures, reject on 3 failures for the 4.50 inch block and accept on 19 failures, reject on 20 failures for the 5.50 inch block assembly."

PAGE 18

* 4.5.4.1: Add at the end of paragraph:

"After fuze assembly it is mandatory to re-insert a safety clip in the fuze prior to the transferring of the sample to the test site. If the slider lock is automatically assembled during the fuze assembly operation, then a removal of the slider lock is necessary to re-insert the safety clip."

PAGE 19

4.5.4.5: Delete the last sentence "If the fuze loosens as evidence by gap..." and replace with the following:

"The sample shall be removed from the fixture and inspected for a gap around the studs between the fuze housing and the cover, and between the fuze cover and the grenade cap. A separating force shall be applied manually to the fuze grenade assembly while gaging. If the sample exceeds the gap permitted on the applicable drawing, the sample assembly shall be classed defective".

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4.5.5 and 4.5.5.1: Delete in their entirety and substitute the following:

"4.5.5 Unseating force test of cone swage and test for presence of entrapped powder of body loading assembly.

4.5.5.1 Preparation of sample items and test procedure for unseating force.

The quantity of body loading assemblies shall be cut in two in a plane perpendicular to the longitudinal axis of the body. The cut shall be located at a minimum distance of .7000 inches from the open end. The cut shall be made cleanly through the body wall and explosive material.

The portion of the cut sample containing the swage of the cone shall have any remaining explosive material cleaned away and then shall be mounted in an approved fixture. The load as specified on the applicable drawings shall be applied to the cone. If the cone is dislodged from its swaged position in the body loading assembly by the specified load, that body loading assembly shall be classed defective."

PAGE 20

4.5.5.2 thru 4.5.5.5: Delete in their entirety and substitute the following:

"4.5.5.2 Test for presence of entrapped powder in body loading assembly. The test may be accomplished by one of the following procedures.

* a. Alternate 1. The quantity of body loading assemblies shall be cut in a plane perpendicular to the axis of the body at a minimum distance of .700 inches from the open end. The cut shall be made cleanly through the body wall, cone and filler material.

The ring containing the base end of the cone shall then be cut into two semi-circular halves so that the cone can be lifted out. An alternative to the above method is to cut the cone into two semi-circular halves leaving the base end of the body in ring form. The cone halves shall then be removed but without loss of any entrapped powder.

The two halves of the cone shall be lifted out and the body shall be visually examined for evidence of caked powder entrapped in the crimping groove (see 4.5.5.4).

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* b. Alternate 2. The quantity of body loading assemblies shall be cut on the longitudinal axis in two halves. The cut shall be made cleanly through the body wall, the explosive material, and cone. The sectioned cone in each half shall be lifted out and the body shall be visually examined for evidence of caked powder entrapped in the crimping groove (see 4.5.5.4).

4.5.5.3 Combined testing of force test and test for entrapped powder (alternative method). The force test and test for entrapped powder may be combined, provided the contractor uses powder nests or equivalent loading aids to reduce the occurrence of entrapped powder during grenade body loading. To use this alternative the contractor must, prior to use, request approval from the procuring agency and furnish documentation of the method planned to reduce the possibility of entrapped powder. For this combined test the same samples are used for both the force test and entrapped powder test. After having passed the 2000 lb. force requirement, the load shall be increased until the cone is dislodged from the grenade body. The grenade body shall be examined for evidence of powder that had been entrapped in the crimping groove (see 4.5.5.4).

4.5.5.4 Reject criteria of the test for presence of entrapped powder in body loading assembly. The body loading assembly shall be cause for the rejection, if the test reveals any one of the following:

a. The explosive material fills the crimping groove between upper and lower swaging ridge in excess of 30 degrees of the circumference and it is either continuous or additive.

b. The presence of explosive material in the crimping groove is in excess of 90 degrees of the circumference and it is either continuous or additive.

4.5.5.5 Disposition. If any body loading assembly fails to comply with any of these tests the production quantity represented by the test sample shall be rejected."

PAGE 22

* Delete paragraph 4.5.6.2 and substitute:

"4.5.6.2 Modified standoff.

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4.5.6.2.1 Cone configuration. Each body loading assembly to be subjected to this test shall be set into a tube or other approved holding device such that the normal standoff distance is increased by the modified standoff distance (see 3.3). The block, of material and height in accordance with 3.3, shall measure 2 1/2 + 1/16 inches wide and a minimum of 18 inches long. The block shall be placed on a firm horizontal surface with 2 1/2 inch width uppermost and the body loading assembly (with increased standoff) placed on it so the centerline of the cone is 2 inches from the end (End A). Each subsequent shot is to be located a distance of 2 inches from the centerline of the previous hole and the centerline of the last (eighth) shot for that surface shall be located a minimum of 2 inches from the opposite end. The block shall then be turned over (180°) and the centerline of the cone shall be located 3 inches from the end (End A). Each subsequent shot is to be located a distance of 2 inches from the centerline of the previous hole, and the centerline of the last (fifteenth) shot shall be located a minimum of 3 inches from the opposite end. The depth of shots failing to penetrate the entire block height shall be measured with a depth gage or other probe. Should any measurement fail to comply with the specified minimum depth requirement, and lot acceptance is dependent upon the result of that shot, the depth shall again be measured, using radiographic means.

4.5.6.2.2 Liner configuration. Each body loading assembly to be subjected to this test shall be set into a tube or other approved holding device such that the normal standoff distance is increased by the modified standoff distance (see 3.3). The test blocks, of material and height in accordance with 3.3, shall measure 2 1/2 + 1/16 inches wide and a minimum of 18 inches long. The assembly interface surface of the blocks shall be free of burrs and high spots which would prevent a good flush contact between the surfaces. The block assembly shall be placed on a firm horizontal surface (thinner block on bottom) with 2 1/2 inch width uppermost and the body loading assembly with increased standoff placed on it so the centerline of the liner is 2 inches from the end (End A). Each subsequent shot is to be located a distance of 2 inches from the centerline of the previous hole and the centerline of the last (eighth) shot shall be located a minimum of 2 inches from the opposite end of the block. The top block shall then be turned over (180°) and the centerline of the liner be located 3.00 inches from the end (End A). Each subsequent shot is to be located a distance of 2 inches from the centerline of the previous hole, and the centerline of the last (fifteenth)

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shot shall be located a minimum of 3 inches from the opposite end. The depth of shots failing to penetrate the applicable block height shall be measured with a depth gage or other probe. Should any measurement fail to comply with the specified minimum depth requirement, and lot acceptance is dependent upon the result of that shot, the depth shall again be measured, using radiographic means.

NOTE: The above is a recommended test procedure. Requests for the use of alternate procedures shall be submitted to Commander, US ARDC, ATTN: AMSMC-QAR-R(D), Dover, NJ 07801-5001, for approval prior to use."

PAGE 23

4.5.9 and 4.5.10: Add the following new paragraphs:

"4.5.9 Pull test of the slider lock (non-destructive). Each sample slider lock shall be assembled to a fixture simulating a maximum dimension fuze housing and pull tested until the slider lock releases. The load shall be applied to the top of the lock in a direction away from the simulated housing. Any slider lock failing to meet the force requirement on the applicable drawing shall be classed defective.

4.5.10 Hardness test of the slider lock (non-destructive). This test shall be performed in accordannce with ASTM-E-18 using the standard penetrator for the specified hardness scale. Readings shall be taken on the top surface of the slider lock. Any slider lock failing to meet the applicable hardness requirement shall be classed defective."

5: Delete in its entirety and substitute the following:

"5. PACKAGING (This section is not applicable to intraplant usage).

5.1 Packaging. Level A (not intended for long term storage) and Level B: Packaging shall be in accordance with drawing 9349336.

5.2 Packing. Level A (not intended for long term storage) and Level B: Packing shall be in accordance with drawing 9349337.

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5.3 Marking. Marking of fiberboard box and bag shall be in accordance with drawing 9349336. Outer box shall be marked in accordance with drawing 9349337."

6.4: Add the paragraph column, "4.5.9 and 4.5.10" for Quality Inspection Test Demonstration and Evaluation Report."

PAGE 25

* Add new paragraph 6.12:

"6.12 This specification has provisions for two (2) shape-charge configuration-original and improved-for LAP. The original has a plain straight wall cone and the improved has a trumpet-shape liner. If the liner is used in LAP, Dwg. 9349371 shall be referenced in place of Dwg. 9215372 and wherever the cone is noted, change to "liner", except use applicable subparagraph in paragraph 3.3.1, 3.3.2, 4.3.3, 4.5.6.2 and CD 110 in paragraph 4.4.2.4."

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